Trainings

General Information

This procedure provides a general overview of testing an engine air intake and exhaust system for leaks. If a Smoke Machine is being used, a quick reference guide is located on the smoke machine. This procedure should be used when troubleshooting intake boost leaks for low power, poor fuel economy, and for aftertreatment complaints of frequent regeneration or the smell of exhaust during regeneration.

The smoke machine references in this procedure are based on the Redline HD Power Smoke machine and Uptime plugs or an equivalent smoke machine and plugs. Refer to the smoke machine and/or service plug manufacturer service information for proper use and operation.

Select Service Tools

Recommended Cummins® Service Tools

- Charge air cooler leak test kit, Part Number 4919085

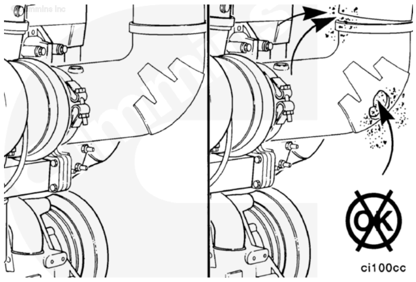
Additional Service Items

- Smoke machine capable of maintaining 34 kPa [5 psi] air pressure
- Air pressure regulator capable of limiting shop air to 34 kPa [5 psi]
- Adapter fitting, male M14X1.5 metric straight thread o-ring, female ¼-18 NPT
- Snoop® liquid leak detector or dish soap/water solution
- Incandescent flashlight
- Mirror
- Thin plastic trash bag
- Gaffer's tape
- Inflatable air bladder(s)

Initial Check

⚠ **CAUTION** ⚠

Engine intake air must be filtered to prevent dirt and debris from entering the engine. If the intake air piping is damaged or loose, unfiltered air will enter the engine and cause premature wear.



Inspect for loose clamps or damage between the following:

- Intake air piping
- Air cleaner
- Turbocharger
- Charge air cooler
- Intake throttle actuator, if equipped
- Air intake connection adapter, if equipped
- Air shutoff valve, if equipped
- Intake manifold
- Air compressor intake tube.

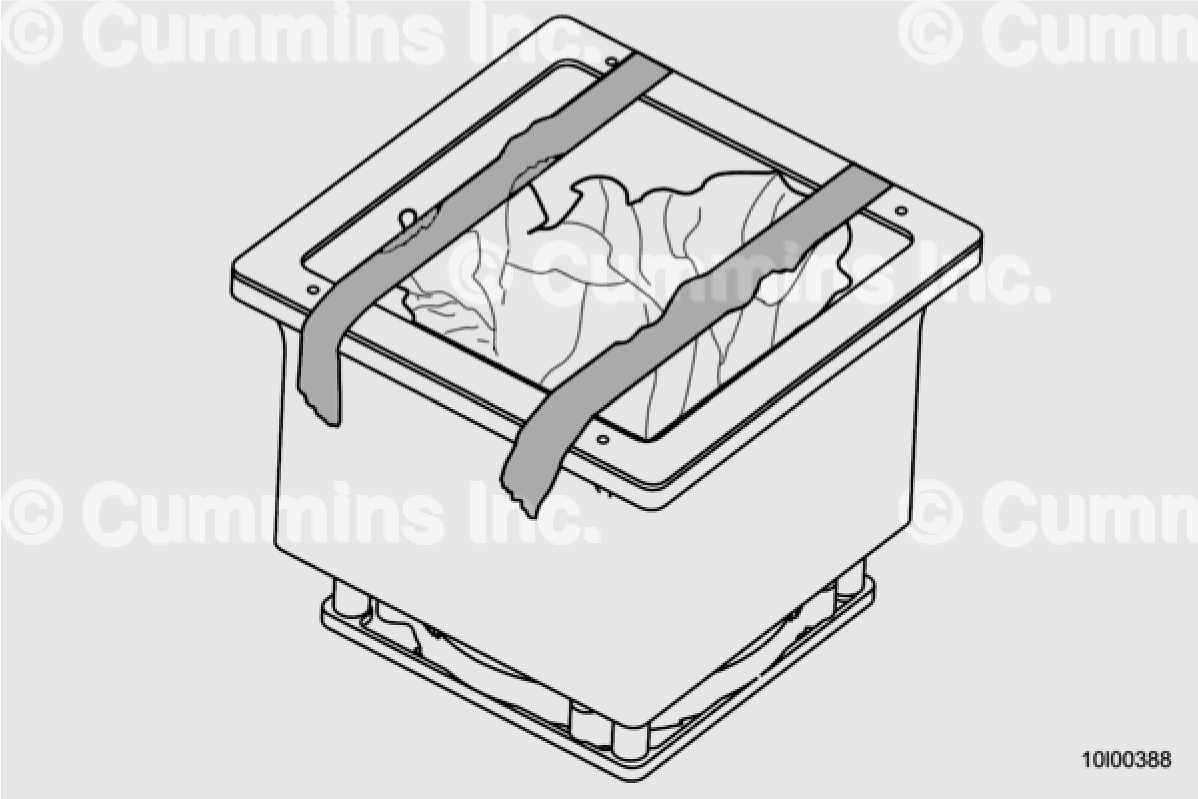
Check for corrosion of the intake system piping under the clamps and hoses. Corrosion can allow corrosive products and dirt to enter the intake system. Disassemble and clean as required.

Replace any damaged pipes, and tighten any loose clamps.

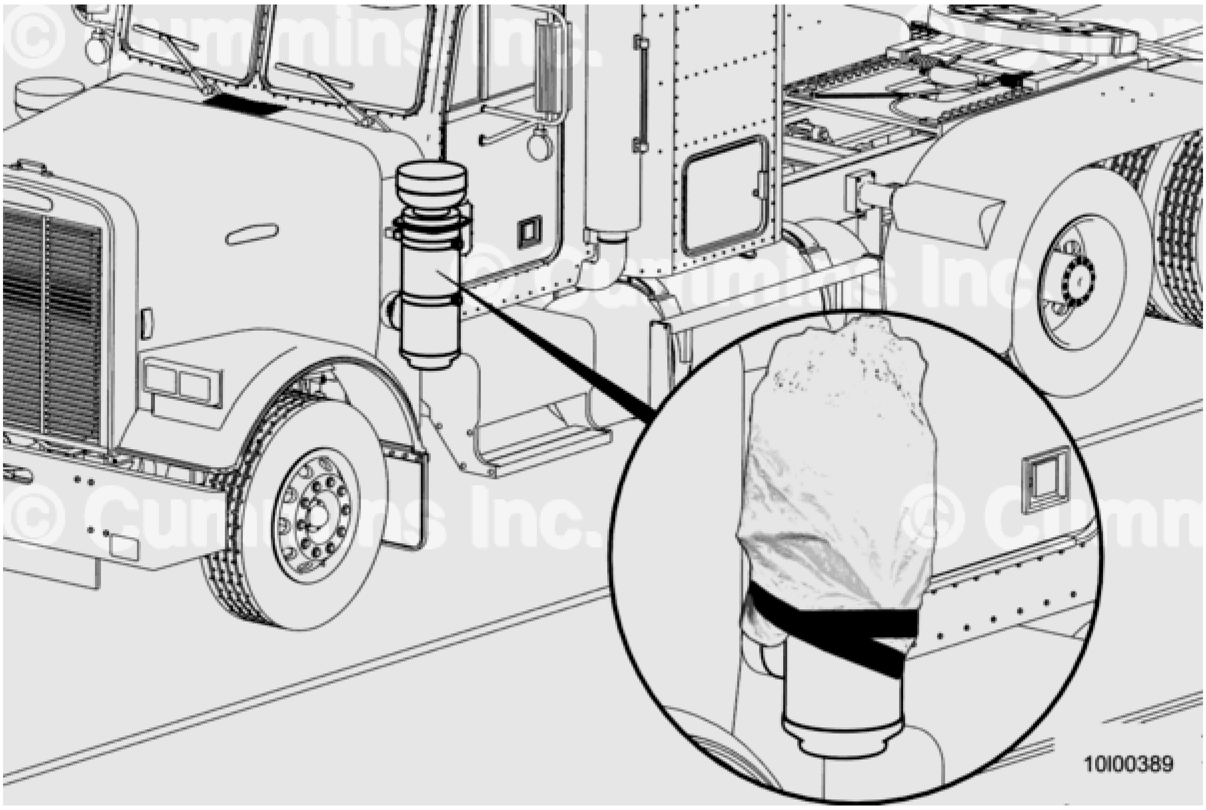
Air Intake System

1.Verify engine is off and shop air 345 to 1206 kPa [50 to 175 psi] is readily available.

2. Use a light duty trash bag and gaffer's tape or utilize inflatable bladder(s) to block off the air filter inlet to allow pressurization of the intake system.

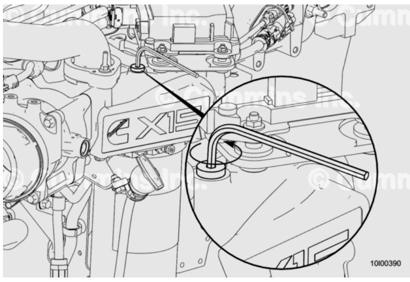


Under hood type air cleaner

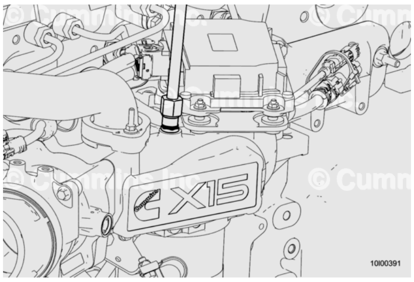


Outside hood type air cleaner

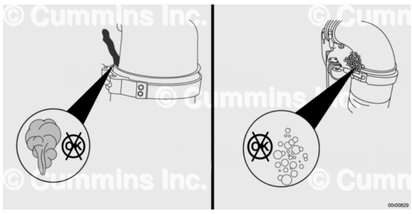
3. Remove the plug from the intake manifold using a 6 mm [
0.236 in] hex key wrench.



4. Install the male M14X1.5 metric straight thread o-ring, female ¼-18 NPT fitting into the intake manifold in place of the plug.



5. Connect the smoke machine or regulated shop air to the intake manifold using the M14X1.5 metric straight thread fitting.



6. Pressurize the air intake system to 34 kPa [5 psi] using the regulated shop air or smoke machine. Do **not** over-pressurize the system. Allow 5 minutes for the intake system to fully pressurize.

7. Look for signs of smoke if using a smoke machine. Using an incandescent flashlight and mirror can help identify smoke leaks.

8. Inspect all air intake system components, connections, and joints for intake air system leaks. Pay special attention to:

a. Turbocharger to the air compressor cylinder head piping
b. Intake piping from the air filter housing to the turbocharger inlet
c. Air filter housing
   1. It is common to see some intake air system leaks where the plastic/tape is sealing the turbocharger inlet.
   2. It is common to see some intake air system leaks on the dirty side of the air filter/box.

9. If no air leaks are found, or if performing without a smoke machine, add Snoop® or soapy water solution to all components and joints of the intake air system and continue to pressurize the system to 34 kPa [5 psi].

10. Inspect all intake system components, connections, and joints for intake air system leaks including:

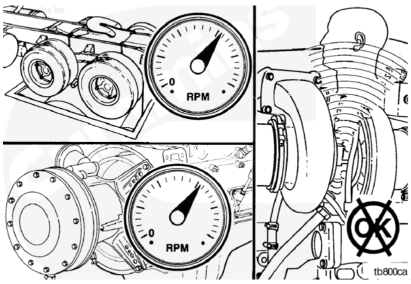
a. Turbocharger to the air compressor cylinder head
b. Intake piping from the air filter housing to the turbocharger inlet
c. Air filter housing
   1. It is common to see some intake air system leaks where the plastic/tape is sealing the turbocharger inlet.
   2. It is common to see some intake air system leaks on the dirty side of the air filter/box.

11. Take pictures and/ or videos of any intake air system leaks seen.

12. See Service Bulletin, High Blowby and Lubricating Oil Consumption Caused by Dirt and Dust Ingestion, Bulletin 5613318 (/qs3/pubsys2/xml/en/bulletin/5613318.html) if intake air system leaks are found.

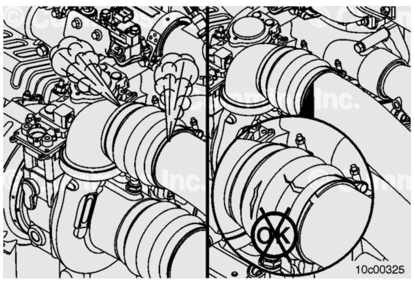
Charge Air and Turbocharger Exhaust Leak Check

Excessive back pressure can cause exhaust leaks. Verify the exhaust back pressure is within specified limits. Refer to Procedure 011-009 in Section 11. (/qs3/pubsys2/xml/en/procedures/377/377-011-009.html) Operate the engine at full throttle and maximum load, and check for air leaks. Listen for a whistling noise caused by high-pressure air leaks.



⚠ CAUTION ⚠

Do not use air tools to remove or install the nut on the V-band clamp. Use of these tools can seriously damage the threads or the bolt and cause the clamp to not be able to be used.



The noise can be caused by an air leak from the following:

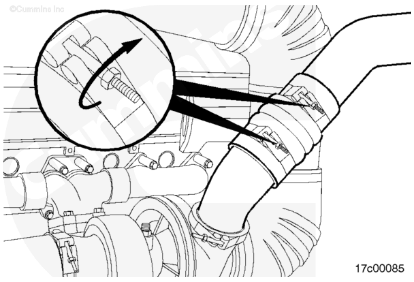
- Turbocharger-to-charge air cooler elbow connection.
  - Inspect the connection and o-ring for damage.
  - Tighten the V-band clamps.

Torque Value: 9 n•m [80 in-lb]

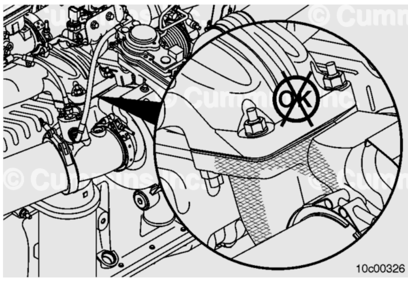
- Any charge air cooler piping or connecting hose.
  - Inspect the hose and piping for damage.

- Tighten the hose clamps.

Torque Value: 9 n•m [80 in-lb]



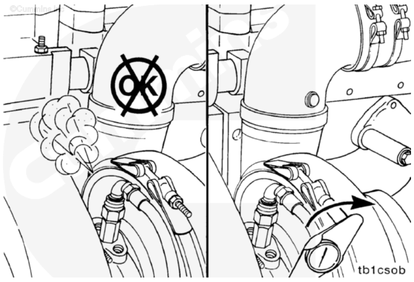
- Check soot streaks from the turbocharger-to-exhaust manifold mounting gasket.
  - Inspect for signs of leakage, such as soot streaks which may appear yellowish, brown, or black. If signs of leakage are present, replace the turbocharger mounting gasket. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/377/377-010-033.html)



- Compressor housing sealing surface air leak.
  - Tighten the V-band.

Torque Value: 9 n•m [80 in-lb]

- Check for an air leak.
  - If an air leak is still present, remove and replace the turbocharger. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/377/377-010-033.html)

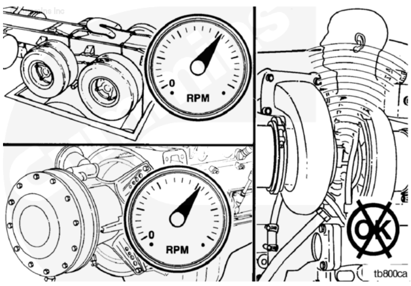


Exhaust System

Without Smoke Machine

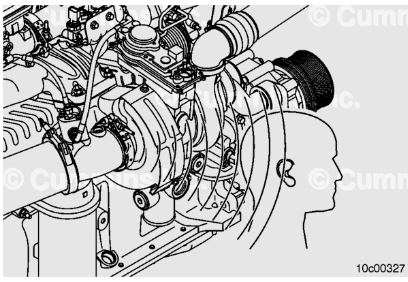
Excessive back pressure can cause exhaust leaks. Verify the exhaust back pressure is within specified limits. Refer to Procedure 011-009 (/qs3/pubsys2/xml/en/procedures/377/377-011-009.html) in Section 11.

Operate the engine at full throttle and maximum load, and check for air leaks. Listen for a whistling noise caused by high-pressure air leaks.



⚠ CAUTION ⚠

Removing the turbine housing V-band tamper proofing on any variable geometry turbocharger will void the Cummins Inc. warranty coverage.



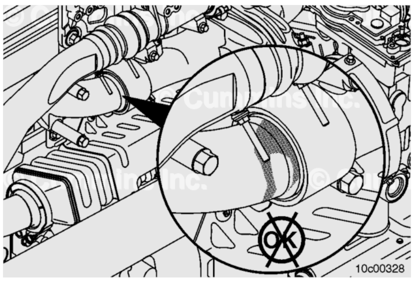
- Variable geometry turbocharger turbine housing sealing surface air leak.

For variable geometry turbochargers, the turbine V-band nut has a tamper proof clamp. Therefore, the torque specification for the turbine V-band can **not** be checked.

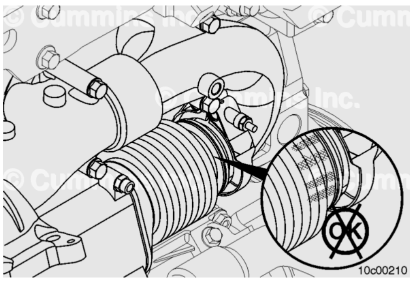
The purpose of the tamper proofing is to prevent disturbing this joint. Any loosening of the V-band or adjusting of the orientation of the turbine housing will affect the sliding nozzle clearances and will damage the turbocharger, which will result in poor performance.

Indications of leakage are soot streaks from the joint and/or an audible whistle from the engine. The turbocharger **must** be replaced if there are any indications of leakage from the turbine housing V-band joint.

- Check for soot streaks from the exhaust manifold slip joints and exhaust manifold gasket.
  - Inspect for signs of leakage, such as soot streaks which may appear yellowish, brown, or black.
  - If signs of leakage are present, replace the slip joints or exhaust manifold gaskets. Refer to Procedure 011-007 in Section 11. (/qs3/pubsys2/xml/en/procedures/377/377-011-007.html)

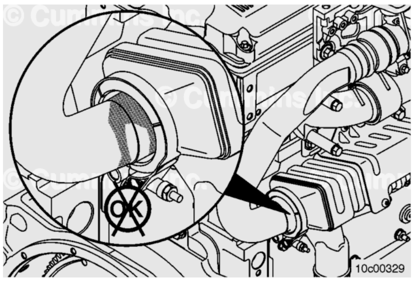


- Check for leakage from the exhaust gas recirculation (EGR) cooler bellows.
  - Inspect for signs of leakage such as soot streaks.
  - If signs of leakage are present, replace the EGR cooler bellows or seals. Refer to Procedure 011-024 in Section 11. (/qs3/pubsys2/xml/en/procedures/377/377-011-024.html)

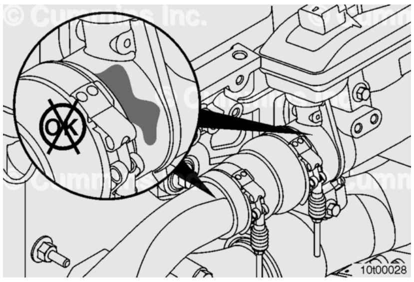


- Check for soot streaks from the rear EGR connection tube.

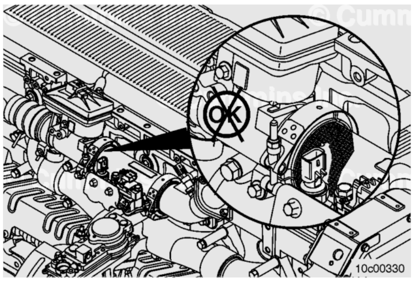
- Tighten the V-band clamp. Refer to Procedure 011-069 in Section 11. (/qs3/pubsys2/xml/en/procedures/377/377-011-069.html)
- Check for an exhaust leak.
- If an exhaust leak is still present, replace the rear EGR connection tube o-ring.



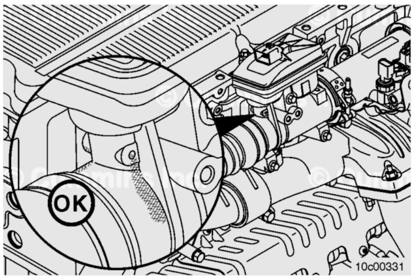
- Check for soot streaks from the EGR hose.
  - If an exhaust leak is present from the EGR hose, tighten the hose clamps. Refer to Procedure 011-069 in Section 11. (/qs3/pubsys2/xml/en/procedures/377/377-011-069.html)



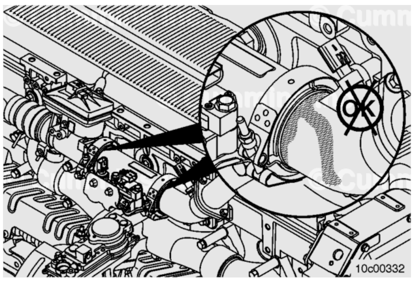
- Check for soot streaks from the EGR valve o-ring.
  - Inspect for signs of leakage such as soot streaks.
  - If signs of leakage are present, replace the EGR valve o-ring. Refer to Procedure 011-022 in Section 11. (/qs3/pubsys2/xml/en/procedures/377/377-011-022.html)



- Check for leakage from the EGR valve weep hole.
  - Soot leakage at the EGR valve weep hole is normal and is expected. Do **not** replace the EGR valve if soot streaks are found at the EGR valve weep hole.

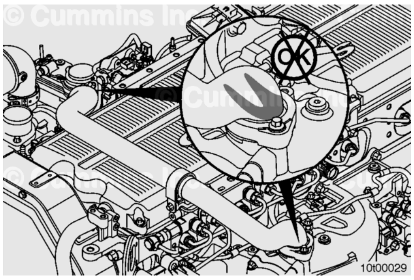


- Check for soot leakage at the EGR mass measurement flow assembly and EGR differential pressure sensor.
  - Inspect for signs of leakage such as soot streaks which may appear yellowish, brown, or black.
  - If signs of leakage are present, replace the EGR mass measurement flow assembly o-rings. Refer to Procedure 011-066 in Section 11. (/qs3/pubsys2/xml/en/procedures/377/377-011-066.html)
  - Replace the EGR differential pressure sensor. Refer to Procedure 019-370 in Section 19.

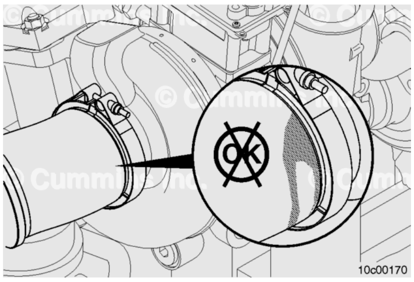


(/qs3/pubsys2/xml/en/procedures/377/377-019-370.html)

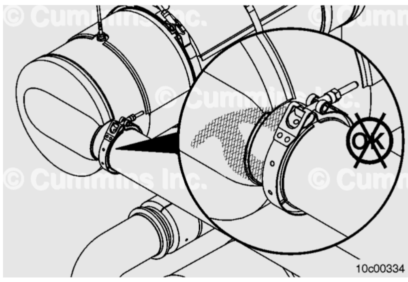
- Check for soot leakage at the exhaust transfer connection elbow and EGR crossover tube.
  - Inspect for signs of leakage such as soot streaks which may appear yellowish, brown, or black.
  - If signs of leakage are present, replace the EGR crossover tube seal carriers. Refer to Procedure 011-070 in Section 11. (/qs3/pubsys2/xml/en/procedures/377/377-011-070-tr.html)



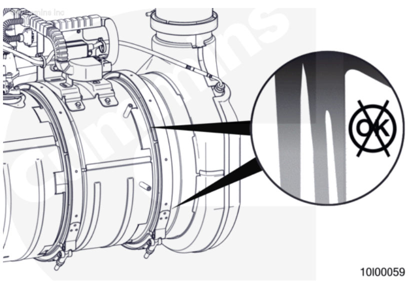
- Check the aftertreatment adapter pipe.
  - Inspect for visible soot streaks at the V-band joint.
  - If soot streaks are found, tighten the V-band clamp. Refer to Procedure 011-043 in Section 11. (/qs3/pubsys2/xml/en/procedures/377/377-011-043.html)
- Operate the engine at full throttle and check for leakage at full throttle condition
  - If a leak is still present, replace the gasket and V-band clamp.



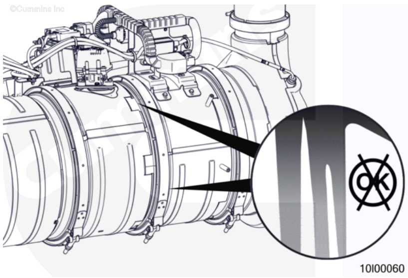
- Check the aftertreatment inlet connection.
  - Inspect for visible soot streaks at the V-band clamp joint. Also inspect for loose or damaged insulation wrapping, if applicable.
  - If soot streaks are found, tighten the V-band clamp. Refer to Procedure 011-041 in Section 11. (/qs3/pubsys2/xml/en/procedures/493/493-011-041.html)
- Operate the engine at full throttle and check for leakage at full throttle condition.
  - If a leak is still present, replace the V-band clamp.



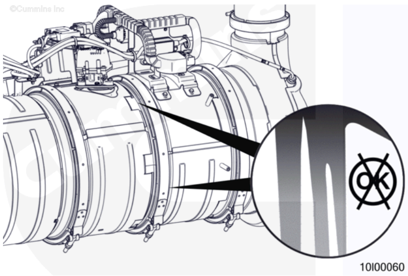
- Check the catalyst to aftertreatment diesel particulate (DPF) filter.
  - Inspect for visible soot streaks at the V-band joint.
  - If soot streaks are found, check for fault codes relating to high differential pressure in the aftertreatment system. If no fault codes are found, tighten the V-band clamp. Refer to Procedure 011-



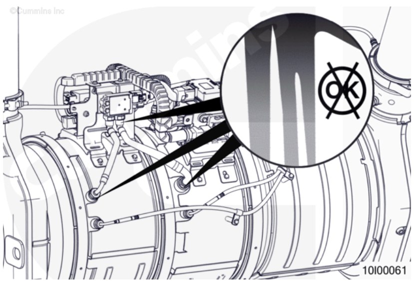
049 in Section 11.
(/qs3/pubsys2/xml/en/procedures/493/493-011-049.html)
- Operate the engine at full throttle and check for leakage at full throttle condition.
    - If a leak is still present, replace the gasket and V-band clamp.



- Check the aftertreatment DPF to outlet connection.
    - Inspect for visible soot streaks at the V-band joint.
    - If soot streaks are found, check for fault codes relating to high differential pressure in the aftertreatment system. If no fault codes are found, tighten the V-band clamp. Refer to Procedure 011-041 in Section 11. (/qs3/pubsys2/xml/en/procedures/493/493-011-041.html)
- Operate the engine at full throttle and check for leakage at full throttle condition.
    - If a leak is still present, replace the gasket and V-band clamp.



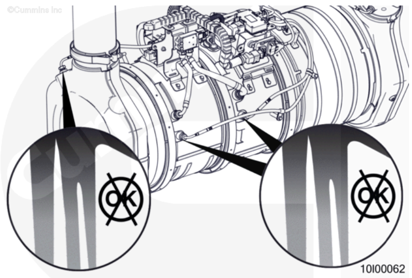
- Check the aftertreatment outlet connection.
    - Inspect for visible soot streaks at the V-band joint.
    - If soot streaks are found, check for fault codes relating to high differential pressure in the aftertreatment system. If no fault codes are found, tighten the V-band clamp. Refer to Procedure 011-041 in Section 11. (/qs3/pubsys2/xml/en/procedures/493/493-011-041.html)
- Operate the engine at full throttle and check for leakage at full throttle condition.
    - If a leak is still present, replace the gasket and V-band clamp.



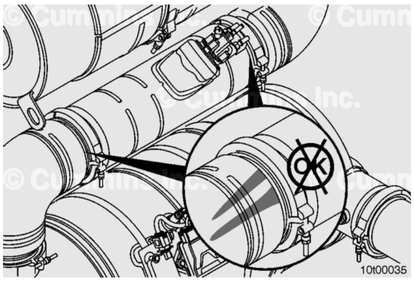
- Check the aftertreatment DPF differential pressure sensor tube connections.
    - Inspect for visible soot streaks at the threaded bosses or hose connections.
    - If soot streaks are found at the threaded boss, loosen the differential pressure sensor tube nut, apply a coating of anti-seize compound, and tighten the differential pressure sensor tube nut. Refer to

Procedure 011-047 in Section 11.
(/qs3/pubsys2/xml/en/procedures/493/493-011-047.html)
    - If soot streaks are found at the hose connections, remove and connect the hose. Refer to Procedure 011-047 in Section 11.
(/qs3/pubsys2/xml/en/procedures/493/493-011-047.html)
- Operate the engine at full throttle and check for leakage at full throttle condition.
    - If a leak is still present at the threaded boss, replace the differential pressure sensor tube. Refer to Procedure 011-047 in Section 11.
(/qs3/pubsys2/xml/en/procedures/493/493-011-047.html)
    - If a leak is still present at the hose connection, replace the differential pressure sensor tube or differential pressure sensor, depending on which component the hose is permanently attached to. Refer to Procedure 011-047 in Section 11.
(/qs3/pubsys2/xml/en/procedures/493/493-011-047.html) Refer to Procedure 019-443 in Section 19.
(/qs3/pubsys2/xml/en/procedures/493/493-019-443.html)
    - If damage resulted in coolant, oil, excessive fuel or excessive black smoke entering the exhaust system, the aftertreatment system **must** be inspected.
        - Use the following procedure to check the aftertreatment diesel particulate filter (DPF). Refer to Procedure 014-016 in Section 14.
(/qs3/pubsys2/xml/en/procedures/493/493-014-016.html)

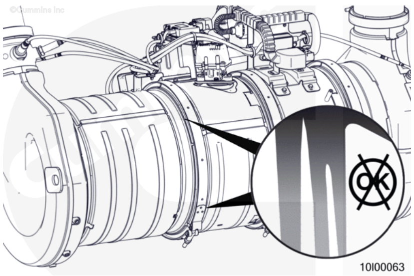
- Check the aftertreatment diesel oxidation catalyst (DOC) intake temperature connection, and the DPF intake and outlet temperature connections.
    - Inspect for visible soot streaks at the threaded bosses.
    - If soot streaks are found at the threaded boss, loosen the differential temperature sensor nut and inspect the threads for damage. If damage is found, replace the sensor. Refer to Procedure 019-449 in Section 19.
(/qs3/pubsys2/xml/en/procedures/493/493-019-449.html)



- If no damage is found, apply a coating of anti-seize compound and tighten the temperature sensor nut. Refer to Procedure 019-449 in Section 19. (/qs3/pubsys2/xml/en/procedures/493/493-019-449.html)
- Operate the engine at full throttle and check for leakage at full throttle condition.
    - If a leak is still present at the threaded boss, replace the temperature sensor. Refer to Procedure 019-449 in Section 19. (/qs3/pubsys2/xml/en/procedures/493/493-019-449.html)



- Check the aftertreatment decomposition pipe and its related components connecting the DPF to the selective catalytic reduction (SCR) catalyst.
    - Inspect for visible soot streaks at the V-band clamp joint. Also inspect for loose or damaged insulation wrapping, if applicable.
    - If soot streaks are found, tighten the V-band clamp. Refer to Procedure 011-062 (/qs3/pubsys2/xml/en/procedures/377/377-011-062.html) in Section 11.
    - Inspect for visible crystallized diesel exhaust fluid (DEF) at the V-band clamp joint and surrounding areas after the DEF dosing valve.
    - If crystallized DEF is found, tighten the V-band clamp. Refer to Procedure 011-062 (/qs3/pubsys2/xml/en/procedures/377/377-011-062.html) in Section 11.
- Operate the engine at full throttle and check for leakage at full throttle condition.
    - If a leak is still present, replace the clamp with a new V-band clamp.

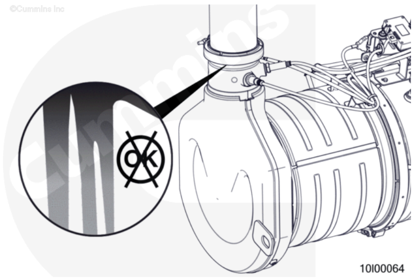


- Check the aftertreatment SCR catalyst inlet connection.
    - Inspect for visible soot streaks at the V-band clamp joint. Also inspect for loose or damaged insulation wrapping, if applicable.
    - If soot streaks are found, tighten the V-band clamp. Refer to Procedure 011-036 in Section 11. (/qs3/pubsys2/xml/en/procedures/493/493-011-036.html)

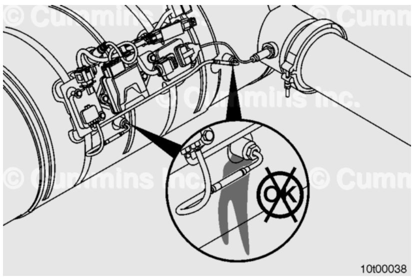
- Inspect for visible crystallized DEF at the V-band clamp joint and surrounding areas.
- If crystallized DEF is found, tighten the V-band clamp. Refer to Procedure 011-036 in Section 11. (/qs3/pubsys2/xml/en/procedures/493/493-011-036.html)

* Operate the engine at full throttle and check for leakage at full throttle condition.
  - If a leak is still present, replace the V-band clamp.

* Check the aftertreatment SCR catalyst outlet connection.
  - Inspect for visible soot streaks at the Torca™ clamp or V-band joint.
  - Inspect the exhaust catalyst for signs of DEF leaks.
  - If soot streaks or signs of DEF leaks are found, tighten the Torca™ clamp or V-band clamp.
* Operate the engine at full throttle and check for leakage at full throttle condition.
  - If a leak is still present, replace the gasket and Torca™ clamp or V-band clamp.

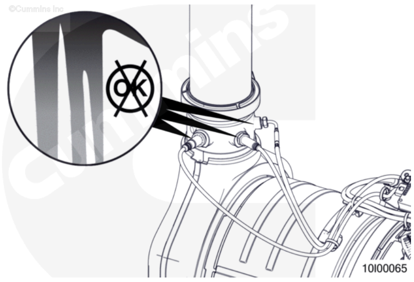


* Check the aftertreatment SCR catalyst inlet and outlet temperature sensors connections.
  - Inspect for visible soot streaks at the threaded bosses.
  - If soot streaks are found at the threaded boss, loosen the temperature sensor nut, apply a coating of anti-seize compound, and tighten the temperature sensor nut. Refer to Procedure 019-449 (/qs3/pubsys2/xml/en/procedures/493/493-019-449.html) in Section 19.
* Operate the engine at full throttle and check for leakage at full throttle condition.
  - If a leak is still present at the threaded boss, replace the temperature sensor. Refer to Procedure 019-449 (/qs3/pubsys2/xml/en/procedures/493/493-019-449.html) in Section 19.



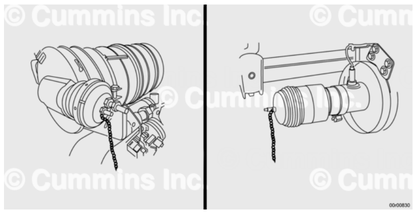
* Check the aftertreatment outlet temperature sensor, NOx sensor, and particulate matter sensor connection.
  - Inspect for visible soot streaks at the threaded boss.
  - If soot streaks are found at the threaded boss, loosen the aftertreatment outlet NOx sensor probe nut, apply a coating of anti-seize compound, and

tighten the aftertreatment outlet NOx sensor probe nut. Refer to Procedure 019-451 in Section 19. (/qs3/pubsys2/xml/en/procedures/493/493-019-451.html)



- Operate the engine at full throttle and check for leakage at full throttle condition.
    - If a leak is still present at the threaded boss, replace the aftertreatment outlet NOX sensor. Refer to Procedure 019-451 in Section 19. (/qs3/pubsys2/xml/en/procedures/493/493-019-451.html)

With Smoke Machine



Do **not** use smoke to test for leaks on aftertreatment.

Use pressurized air and a soapy water bottle to check Aftertreatment for leaks “Bubbles". Smoke will **not** be seen due to Aftertreatment design.

Find a place in the exhaust system as close to, but before the DOC where the pipe can be easily separated. Install the plug into the inlet pipe with the pass-through passage open to allow shop air into the Exhaust and Aftertreatment system. Refer to service plug manufacturer service information.

Find a place in the exhaust system as close to, but after the SCR outlet where the pipe can be easily separated. Install the plug into the pipe outlet with the pass-through passage closed to keep shop air from exiting the pipe. Refer to service plug manufacturer service information.

Note : Cleaning the inside of the exhaust pipe with contact cleaner and a clean shop towel will improve sealing and reduce the possibility of the plug coming out of the pipe.

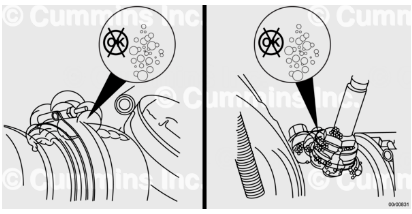
⚠ CAUTION ⚠

Refer to smoke machine or service plug manufacturer service information.

⚠ WARNING ⚠

Do not over pressurize the system or exceed 207 kPa [30 psi] (Max).

Note : Uptime plugs have a built-in pressure relief valve to prevent damage and over pressurization.



⚠ WARNING ⚠

To reduce the possibility of injury if either plug blows off during the test, ensure plug is fully inserted into the pipe and secure safety chains on the test plugs to any convenient secured fixture. This test must not be performed without securely fastened safety chains.

Note : Placement further from SCR outlet will optimize connections tested.

Do **not** use smoke on Aftertreatment. Use shop air **only.**

Connect the smoke machine and begin testing. Refer to smoke machine manufacturer service information. Use a spray bottle with soapy water to check for “Bubbles”.

Check the exhaust, V-band clamps, aftertreatment plumbing and sensor mounting points prior to the aftertreatment outlet for leaks and repair as needed. Additional leak locations where soot streaks or cracks may occur are listed elsewhere in this procedure. Leaks will be Visible as “Bubbles. If no leaks are found, increase the pressure and recheck for leaks. If no leaks are found, it may be necessary to increase pressure again and re-check for leaks. Do **not** over-pressurize the system.